## Question ID: 7902bed0

| Assessment | Test | Domain | Skill | Difficulty |
| --- | --- | --- | --- | --- |
| SAT | Math | Advanced Math | Nonlinear functions | Hard |

**Question**

A machine launches a softball from ground level. The softball reaches a maximum height of $51.84$ meters above the ground at $1.8$ seconds and hits the ground at $3.6$ seconds. Which equation represents the height above ground $h$, in meters, of the softball $t$ seconds after it is launched?

**Answer**

A. $h = -t^2 + 3.6$

B. $h = -t^2 + 51.84$

C. $h = -16(t - 1.8)^2 - 3.6$

D. $h = -16(t - 1.8)^2 + 51.84$

**Correct Answer: D**

**Rationale**

Choice D is correct. An equation representing the height above ground $h$, in meters, of a softball $t$ seconds after it is launched by a machine from ground level can be written in the form $h = -a(t - b)^2 + c$, where $a$, $b$, and $c$ are positive constants. In this equation, $b$ represents the time, in seconds, at which the softball reaches its maximum height of $c$ meters above the ground. It's given that this softball reaches a maximum height of $51.84$ meters above the ground at $1.8$ seconds; therefore, $b = 1.8$ and $c = 51.84$. Substituting $1.8$ for $b$ and $51.84$ for $c$ in the equation $h = -a(t - b)^2 + c$ yields $h = -a(t - 1.8)^2 + 51.84$. It's also given that this softball hits the ground at $3.6$ seconds; therefore, $h = 0$ when $t = 3.6$. Substituting $0$ for $h$ and $3.6$ for $t$ in the equation $h = -a(t - 1.8)^2 + 51.84$ yields $0 = -a(3.6 - 1.8)^2 + 51.84$, which is equivalent to $0 = -a(1.8)^2 + 51.84$, or $0 = -3.24a + 51.84$. Adding $3.24a$ to both sides of this equation yields $3.24a = 51.84$. Dividing both sides of this equation by $3.24$ yields $a = 16$. Substituting $16$ for $a$ in the equation $h = -a(t - 1.8)^2 + 51.84$ yields $h = -16(t - 1.8)^2 + 51.84$. Therefore, $h = -16(t - 1.8)^2 + 51.84$ represents the height above ground $h$, in meters, of this softball $t$ seconds after it is launched.

Choice A is incorrect. This equation represents a situation where the maximum height is $3.6$ meters above the ground at $0$ seconds, not $51.84$ meters above the ground at $1.8$ seconds.

Choice B is incorrect. This equation represents a situation where the maximum height is $51.84$ meters above the ground at $0$ seconds, not $1.8$ seconds.

Choice C is incorrect and may result from conceptual or calculation errors.